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Evaluating generative AI-assisted inquiry-based learning and metacognitive scaffolding for enhancing scientific vocabulary, content comprehension, and lab report writing

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ABSTRACT

While generative artificial intelligence (GAI) holds promise for supporting academic language and content learning, its role in CLIL science instruction remains underexplored. This mixed-methods study investigates how two GAI-supported approaches, Inquiry-Based Learning (IBL) and Metacognitive Scaffolding (MS), influence fifth-grade students' vocabulary acquisition, content comprehension, lab report writing, and learning perceptions. Sixty-six students from three Taiwanese elementary schools participated in a 10-week intervention using *ChatGPT 4.0* within either IBL ($n=34$) or MS ($n=32$) frameworks. Data included pre- and post-tests, lab reports, and bilingual survey responses. Both groups made significant gains in vocabulary and content knowledge. However, IBL students demonstrated stronger vocabulary growth and deeper conceptual integration, reflected in more precise terminology and analytical writing. MS students showed clearer procedural structure but tended toward surface-level analysis. While students in both groups viewed *ChatGPT* positively, they also reported occasional difficulty with language complexity. This study contributes empirical evidence and theory-informed design insights for integrating GAI into CLIL instruction. It demonstrates how aligning AI prompts with constructivist (IBL) or metacognitive (MS) principles can promote vocabulary use, scientific reasoning, and reflective writing. Findings support a scalable framework that combines inquiry-driven and metacognitive scaffolds to enhance language-content integration in bilingual science classrooms.

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Generative artificial intelligence (GAI); inquiry-based learning (IBL); metacognitive scaffolding (MS); scientific content comprehension; lab report writing

Introduction

The integration of content and language learning has become a defining feature of twenty-first-century bilingual education. In science classrooms, this integration is crucial for developing scientific literacy, which encompasses understanding scientific concepts, communicating ideas with precision, and applying knowledge in new contexts (Cenoz et al., 2014; Grapin, 2023). Content and Language Integrated Learning (CLIL) provides a framework for these goals, drawing on the 4Cs (Content, Communication, Cognition, and Culture) to combine disciplinary knowledge with academic language development (Coyle et al., 2010). In Taiwan, the 2030 Bilingual Nation Policy positions CLIL as a key strategy for strengthening both English proficiency and STEM readiness (National Development Council, 2021). Within this policy vision, scientific literacy depends on students' ability to acquire and use scientific vocabulary, build conceptual understanding, and communicate findings through structured lab report writing, all while engaging in cognitively demanding inquiry tasks (Xin & Lo, 2025).

However, the dual demands of content and language learning remain a persistent challenge in Taiwanese CLIL science classrooms. For instance, students often struggle to use academic vocabulary meaningfully, limiting their depth of understanding and engagement (Lo et al., 2023). Structured lab report writing is rare at the elementary level, which leaves students with few opportunities to document scientific reasoning in English (Huang, 2021). Moreover, teachers face heavy preparation loads, a shortage of CLIL-specific materials, and limited time to design lessons that integrate inquiry with explicit language support (Tsou, 2021). Without sustained scaffolding, students may resort to rote memorization and miss opportunities to develop higher-

order thinking and authentic scientific communication. These barriers point to the need for scalable, innovative supports that address both the linguistic and cognitive dimensions of CLIL science learning.

Generative Artificial Intelligence (GAI) tools such as *ChatGPT* offer one such possibility. GAI can create interactive, dialogic environments where students pose questions, receive targeted feedback, and refine their reasoning in real time (Dermeval et al., 2018; Yeh, 2024). However, unstructured use of AI risks producing responses that are inaccurate, overly complex, or misaligned with curricular goals. To make GAI effective in CLIL science education, it must be embedded within pedagogically grounded frameworks. This study focuses on two such frameworks: Inquiry-Based Learning (IBL), structured around the 5E model (Bybee et al., 2006; Pedaste et al., 2015), which promotes conceptual exploration, hypothesis testing, and application; and Metacognitive Scaffolding (MS), which develops planning, monitoring, and reflection skills essential for self-regulated learning (Schraw & Moshman, 1995; Zohar & David, 2008). Both have shown promise in enhancing content understanding and academic language, but their relative strengths in GAI-supported CLIL classrooms remain underexplored.

Although prior research has examined AI-supported inquiry or reflective scaffolding separately, few studies have compared their effects on elementary students' scientific vocabulary acquisition, content comprehension, and lab report writing in CLIL science contexts (Nikula, 2015; Pappa, 2021). This study addresses that gap by investigating the differential impact of GAI-assisted IBL and MS, and by examining how these approaches shape students' perceptions of their own learning.

Research questions

1. What are the effects of GAI-assisted IBL and MS on scientific vocabulary acquisition and content comprehension in CLIL science?
2. How do these improvements influence lab report writing, particularly in clarity of hypotheses, data accuracy, depth of analysis, and use of scientific terminology?
3. How do students perceive the effectiveness of GAI-assisted IBL versus MS in supporting vocabulary acquisition, content comprehension, lab report writing, engagement, and motivation?

By addressing these questions, this study contributes to theory, practice, and policy. It provides empirical evidence on how GAI can be embedded in inquiry and metacognitive frameworks to create interactive learning environments that strengthen bilingual scientific literacy. Practically, it offers strategies for teachers to integrate AI tools into CLIL science in ways that are scalable, sustainable, and aligned with Taiwan's bilingual education goals.

Literature review

Inquiry-based learning in GAI-supported interactive environments

IBL is rooted in constructivist theory, where learners actively pose questions, investigate problems, and build understanding through authentic engagement (Vygotsky, 1978). The 5E model (Engage, Explore, Explain, Elaborate, Evaluate) offers a practical structure for scaffolding this process (Bybee, 2015). In CLIL science, this structure supports both content mastery and the purposeful use of academic language. When integrated with generative AI tools such as *ChatGPT*, IBL can become an interactive learning environment where students co-construct meaning with both peers and AI. In the Explore phase, for instance, *ChatGPT* can pose "what if" scenarios or clarify unfamiliar terms to prompt students to articulate reasoning and refine vocabulary (Hutchinson, 2024; Yeh, 2024). In Elaborate, it can present novel, real-world situations that require applying concepts, while in Evaluate, it can guide reflective dialogue about misconceptions or experimental outcomes (Moundridou et al., 2024; Smith-Keiling et al., 2018). This kind of iterative, dialogic exchange reinforces the relationship between inquiry, language use, and deeper conceptual understanding. In this study, these principles directly informed the *ChatGPT* prompt design for the experimental group, aligning each review session with a specific 5E phase. For example, prompts in the Explore phase

encouraged hypothesis generation, while Evaluate-phase prompts scaffolded reflective articulation of content understanding using targeted scientific terms.

Empirical studies support IBL's potential in language-rich STEM contexts. Smith-Keiling et al. (2018) found that structured inquiry improved writing confidence and analytical precision among Asian ELLs in lab settings. Yeh (2024) reported vocabulary gains and enhanced communicative competence in GAI-assisted inquiry for EFL learners. However, much of the existing work examines IBL as a standalone approach; fewer studies directly compare it with other structured scaffolding methods such as metacognitive approaches, particularly in AI-mediated CLIL science learning.

Metacognitive scaffolding in GAI-supported interactive environments

MS emphasizes learners' awareness and regulation of their own thinking (Flavell, 1976; Schraw & Moshman, 1995). Rather than following a fixed sequence like IBL, MS supports ongoing processes of planning, monitoring, self-questioning, and evaluating. In CLIL science, these processes help students manage both the linguistic and cognitive demands of scientific inquiry.

When embedded in an AI-supported interactive learning space, MS allows students to receive adaptive prompts that foster self-regulation. For example, during planning, *ChatGPT* might ask, "What do you predict will happen in your experiment, and why?"; during monitoring, it can guide learners to check their reasoning against observations (Liu et al., 2024; Sun et al., 2024). This dialogue encourages intentional vocabulary use and strategic thinking. These metacognitive strategies guided the *ChatGPT* prompt design for the control group. Prompts were structured to follow a reflective progression: planning goals, self-questioning during review, and evaluating content understanding, which support students' internal regulation of language and content integration.

Studies confirm the effectiveness of metacognitive scaffolding in improving vocabulary acquisition, content comprehension, and scientific writing. For example, Kim et al. (2022) found that GAI-assisted self-monitoring prompts enhanced understanding of complex scientific concepts and reported that metacognitive scaffolding enhanced students' ability to evaluate their explanations and refine scientific writing. Similarly, Chen (2024) compared generative AI chatbots (GACs) with traditional corrective feedback and found that such AI-assisted metacognitive prompts improved vocabulary retention and promoted metalinguistic awareness. Liu et al. (2024) investigated conversational agents (CAs) with knowledge and emotional scaffolding, revealing that dual scaffolding approaches improved knowledge retention and application through self-regulation and engagement. These findings illustrate how GAI-powered MS supports critical thinking, deeper learning, and personalized support, bridging the gap between theory and practice.

While MS is effective for guiding reflection and consolidating knowledge, its less immersive nature compared to IBL may limit opportunities for spontaneous vocabulary use or higher-order conceptual exploration. This distinction reveals the need to examine how these approaches compare when integrated with GAI in CLIL science environments. Such scaffolds are particularly important in bilingual science classrooms, where students must simultaneously process content knowledge and academic English.

Lab report writing as a core component of scientific literacy

Lab report writing is central to CLIL science because it integrates conceptual understanding with academic communication. A typical report includes a hypothesis, methods, results, analysis, and reflection, which each requires precise scientific vocabulary and logical coherence (Grapin, 2023; Törmälä & Kulju, 2023). This process not only documents inquiry outcomes but also supports the development of scientific reasoning. IBL strengthens lab report writing by embedding writing tasks within the inquiry cycle, giving students repeated opportunities to articulate hypotheses, justify interpretations, and link results to theory (Xin & Lo, 2025). MS complements this by helping students plan the structure of their reports, monitor clarity, and evaluate the use of terminology (Hughes & Madrid, 2020; Roca de Larios et al., 2022). In this intervention, MS-driven *ChatGPT* prompts were designed to support this process by asking students to set intentions, monitor their progress, and evaluate the clarity and accuracy of their writing.

Research exhibits the potential of IBL and MS to transform lab report writing in EFL/ESL science contexts. For instance, Stephenson and Sadler-McKnight (2016) demonstrated how the Science Writing Heuristic

(SWH), which emphasizes inquiry, collaboration, and reflection, enhanced students' critical thinking and writing but required linguistic support. Smith-Keiling et al. (2018) extended similar research to Asian ELLs in a biochemistry lab and found that inquiry-based strategies and peer evaluations boosted writing confidence and engagement, though grammatical challenges remained. Kim et al. (2022) examined an GAI-supported scaffolding approach for graduate students, showing that natural language processing (NLP) tools provided expert-level examples and personalized feedback, leading to improved coherence and argumentation.

However, most of this research is in tertiary settings, leaving a gap in understanding how these strategies work for younger CLIL learners still developing foundational writing skills.

Rationale for the present study

Both IBL and MS can be enhanced through generative AI to create rich, interactive learning environments that support the dual goals of CLIL science: deepening conceptual understanding and developing academic English. IBL offers a structured path for inquiry and concept application, while MS provides adaptive prompts for reflection and self-regulation. However, their comparative strengths in fostering scientific vocabulary, content comprehension, and lab report writing, especially for elementary CLIL learners, remain underexplored. The theoretical foundations of IBL and MS were operationalized through carefully structured *ChatGPT* prompts to enable the study to test how each scaffolding approach supports vocabulary development, conceptual understanding, and academic writing. This study addresses this gap by systematically comparing GAI-supported IBL and MS in Taiwanese CLIL science classrooms. In doing so, it examines not only their impact on learning outcomes but also students' perceptions of how these interactive, AI-mediated environments shape their engagement, confidence, and scientific communication skills.

Methodology

Research design and participants

This study employed a 10-week, quasi-experimental, mixed-methods design to examine the effects of two AI-driven instructional approaches, IBL and MS, on scientific vocabulary acquisition, content comprehension, and lab report writing in CLIL science education. A quasi-experimental design was chosen due to logistical constraints that prevented random assignment (Best & Kahn, 2006). Instead, two classes were randomly assigned to the Experimental Group (EG) and two to the Comparison Group (CG) within the same CLIL-based English immersion program.

The participants were fifth graders transitioning from soft CLIL (emphasizing language) to hard CLIL (integrating complex scientific concepts and vocabulary) (Lyster & Ballinger, 2011). Having received four years of CLIL instruction, these students were at a developmental stage suited for early intervention in bilingual science writing and conceptual reasoning. Ethical approval was obtained from the Institutional Review Board, and informed consent was secured from Teacher Facilitators (TFs), students, and students' guardians.

The study included 66 fifth-grade students from three Taiwanese public elementary schools, with 34 in the EG (EG1: $n = 17$; EG2: $n = 17$) and 32 in the CG (CG1: $n = 19$; CG2: $n = 13$). A power analysis confirmed that a minimum of 64 participants was required to detect a moderate effect (Cohen's $d = 0.50$, $\alpha = 0.05$, power = 0.80; Cohen, 2016), and the final sample size slightly exceeded this threshold. An independent samples t -test on pre-test scores showed no significant differences between groups in vocabulary ($t = 1.435$, $p = 0.156$) or content comprehension ($t = 1.155$, $p = 0.252$), confirming comparability at baseline. Additionally, none of the students had prior experience writing lab reports in Chinese or English.

Teacher variability and pedagogical alignment

To ensure instructional consistency across the four classes, the study involved three experienced TFs. The EGs were taught by a Taiwanese CLIL science teacher with 15 years of experience, near-native English proficiency, and a Master's degree in Teaching English from a U.S. university, where he also completed an elementary school internship. The CGs were led by two U.S. native English-speaking teachers, each with over 10 years

of experience teaching in Taiwan's EFL context. All TFs had been teaching science in the CLIL immersion program for more than five years.

Prior to the intervention, all TFs completed four hours of training, which included familiarization with teaching materials, science experiments, and the structure of *ChatGPT*-supported review sessions. During review sessions, they primarily observed student progress and intervened only to clarify misunderstandings or redirect interaction when AI responses deviated from instructional goals. A shared repository of materials and a private communication channel ensured coordination throughout the implementation period.

Instructional timeline and activities

The CLIL science course met twice weekly for 40-minute sessions over 10 weeks. To enable valid comparisons between the IBL and MS frameworks, instructional and assessment activities were carefully sequenced as follows:

- **Week 1:** Students completed pre-tests on scientific vocabulary and content comprehension. Teachers also introduced *ChatGPT* protocols and prompt use to prepare students for AI-assisted review tasks.
- **Weeks 2-8:** Five physical science units (e.g. gravitational force, compression, friction) were taught. Each unit included:
 - A warm-up or concept review (20-25 min),
 - Core instruction with textbook and PPT scaffolds (30 min),
 - A 20-minute *ChatGPT*-supported review aligned with either IBL or MS approaches. These sessions targeted vocabulary and content consolidation, but not writing support.
- **Weeks 4, 6, 9:** After every second unit, students conducted hands-on experiments and submitted individual lab reports based on their observations and reasoning, without teacher or AI input.
- **Week 10:** Post-tests and a bilingual perception survey were administered.

This structure ensured balanced exposure across groups while allowing to isolate the instructional effects of the IBL and MS frameworks on science learning and writing development.

Intervention

Integration of ChatGPT into CLIL review sessions

To support students' development of scientific vocabulary and conceptual understanding, *ChatGPT 4.0* was embedded in structured review sessions aligned with either an IBL or MS framework. All prompts were pre-scripted to align with lesson objectives, target vocabulary, and students' English proficiency (G3-G4 EFL level). These scripts were standardized across schools and pre-validated by the TFs to ensure content fidelity and age-appropriate language. The TFs monitored all *ChatGPT* sessions and provided redirection as needed. To improve transparency and reproducibility, Appendix A provides the complete prompt templates for both conditions, including example questions and tone guidelines. [Figures 1–4](#) below illustrate how these prompts were enacted in classroom settings through student-*ChatGPT* interactions.

EG: inquiry-based learning (5E framework)

Grounded in the IBL framework and the 5E instructional model (Bybee, 2015), *ChatGPT* facilitated review sessions based on the 5E model to reinforce understanding of physical science concepts (e.g. forces, friction, gravity). Students worked in pairs on iPads and engaged with prompts that scaffolded inquiry beyond textbook content. *ChatGPT* encouraged curiosity, vocabulary use, and critical thinking in real time.

- In **Engage**, it initiated discussion with prompts such as, "*Have you ever wondered why some things move when we push or pull them?*" (see [Figure 2](#))
- In **Explore**, it guided predictions and hypotheses, encouraging dialogue and peer reflection.
- In **Explain**, it elicited student-generated examples (e.g. "*Can you think of a time you used a pulling force?*") and provided formative feedback (e.g. "*Great example! Brushing your hair uses a pulling force.*")

Explain phases

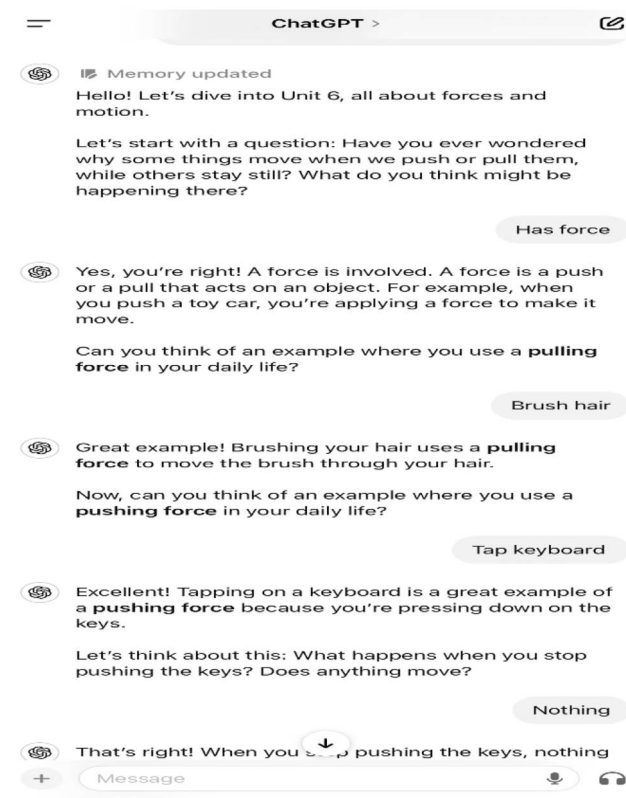


Figure 1. *ChatGPT* in Engage and Explain Phases. Screenshot of a student's interaction with *ChatGPT* during the Engage and Explain phases in an AI-assisted IBL session.

- In **Elaborate**, it challenged students to apply concepts to new contexts, such as, "What kind of force is acting when you press on a sponge-tension, compression, or gravity?"
- In **Evaluate**, *ChatGPT* prompted reflection on learning outcomes (e.g. "What was the most interesting thing you learned today?"). (see [Figure 2](#))

CG: metacognitive scaffolding approach

Informed by metacognitive theory (Flavell, 1976), *ChatGPT* supported review sessions designed to promote reflective learning and self-regulated writing. Students engaged in paired discussions guided by prompts that fostered planning, monitoring, and goal setting.

- In the **Planning** phase, *ChatGPT* encouraged goal articulation (e.g. "What do you think is the main idea we'll explore in today's session about natural forces?"). (see [Figure 3](#))
- In **Monitoring** and **Self-Questioning**, it prompted awareness of learning gaps and deeper inquiry (e.g. "How do you think gravity affects us every day?").
- In **Evaluation**, *ChatGPT* guided students to reflect on progress (e.g. "What's something new you learned today about earthquakes or natural forces?") and set improvement goals (see [Figure 4](#)).

Comparative framework of IBL and MS prompt designs

Grounded in the 5E model, the IBL condition guided students through structured inquiry, enabling them to generate hypotheses, apply scientific concepts, and refine explanations using domain-specific vocabulary. In contrast, the MS condition emphasized metacognitive routines, planning, monitoring, and self-questioning, that supported students' awareness of learning goals and their ability to reflect on progress. While IBL

Evaluate phases

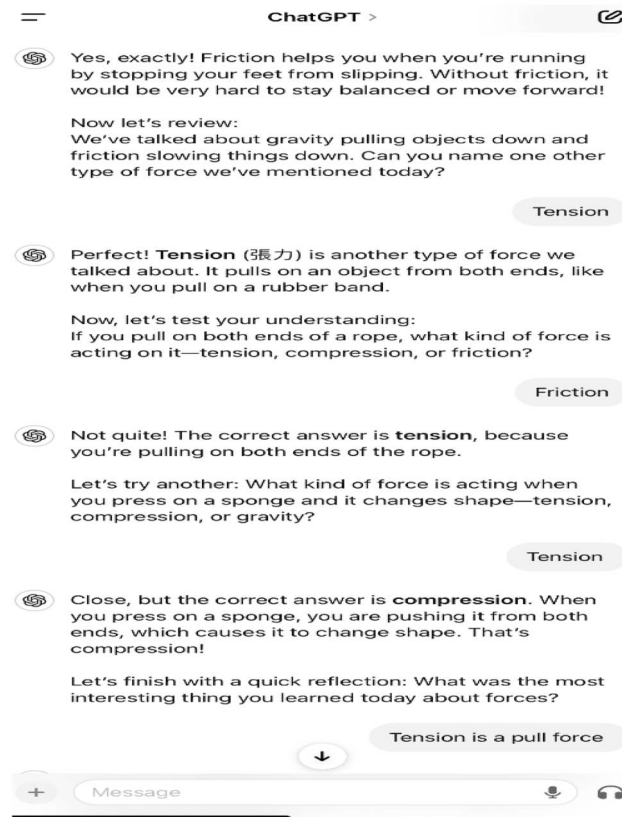


Figure 2. EG's Interaction in Elaborate and Evaluate Phases. Screenshot showing students engaging with *ChatGPT* in the Elaborate and Evaluate phases of the 5E model.

encouraged conceptual exploration and dialogic reasoning, MS provided structure for self-regulation and task ownership. Together, these approaches contribute to a more nuanced understanding of how generative AI can be integrated into CLIL science classrooms to support both linguistic and cognitive development. [Table 1](#) outlines the phases, instructional goals, and sample *ChatGPT* prompts used in each model.

Instruments and analysis

Scientific vocabulary and content comprehension tests (RQ1)

To examine the effects of IBL and MS via *ChatGPT* (RQ1), pre – and post-tests were administered in Week 1 and Week 10. Each test included 20 vocabulary items (matching English definitions with Mandarin translations) and 32 content items (multiple-choice, short-answer, and true/false). To minimize recall bias, post-test items were reordered. All items were aligned with CLIL curriculum objectives and reviewed by three

Table 1. Comparison of IBL and MS Supported by ChatGPT.

IBL/ MS Phase	IBL	MS	Example <i>ChatGPT</i> -Generated Prompts
Engage / Planning	Activate prior knowledge and curiosity	Set learning goals and clarify session focus	"Imagine pushing a box on a slippery floor, what affects it?"/ "What do you want to achieve today?"
Explore / Monitoring	Test hypotheses	Assess and identify difficulties	"What happens if you add weight to an object on a smooth surface?"/ "What confuses you about friction?"
Explain / Self-Questioning	Refine understanding.	Explore "what if" scenarios	"Why do objects move more easily on smooth surfaces? Use friction." / "What would happen if there were no friction?"
Elaborate / Evaluating	Apply knowledge to new scenarios	Reflect on learning and assess progress	"How does friction impact skiing?"/ "What new concept about tension did you learn today?"
Evaluate / Goal-Setting	Reflect on outcomes and assess understanding	Set goals for future learning	"How has your understanding of forces changed?"/ "What will you focus on next to improve?"

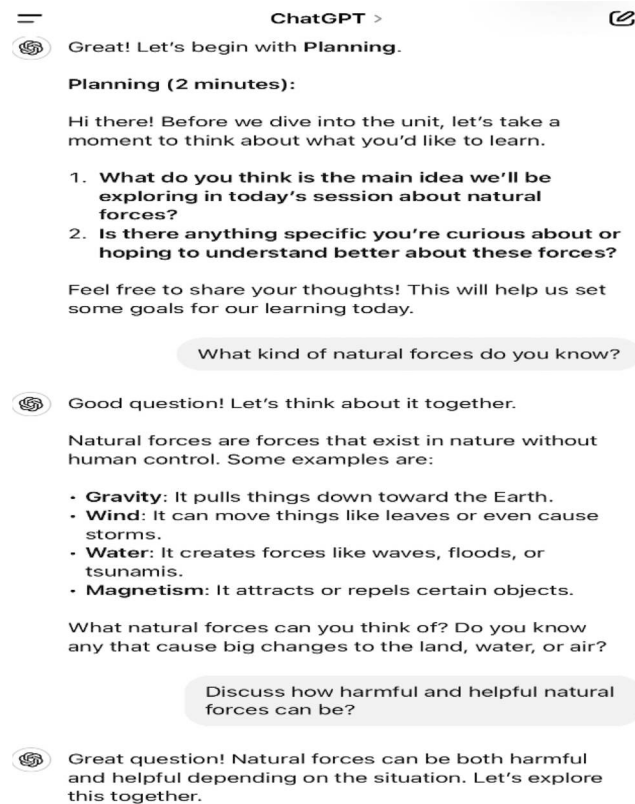


Figure 3. CG's *ChatGPT* Interaction in the Planning Phase. This screenshot shows CG students using *ChatGPT* to set learning goals and activate prior knowledge on natural forces.

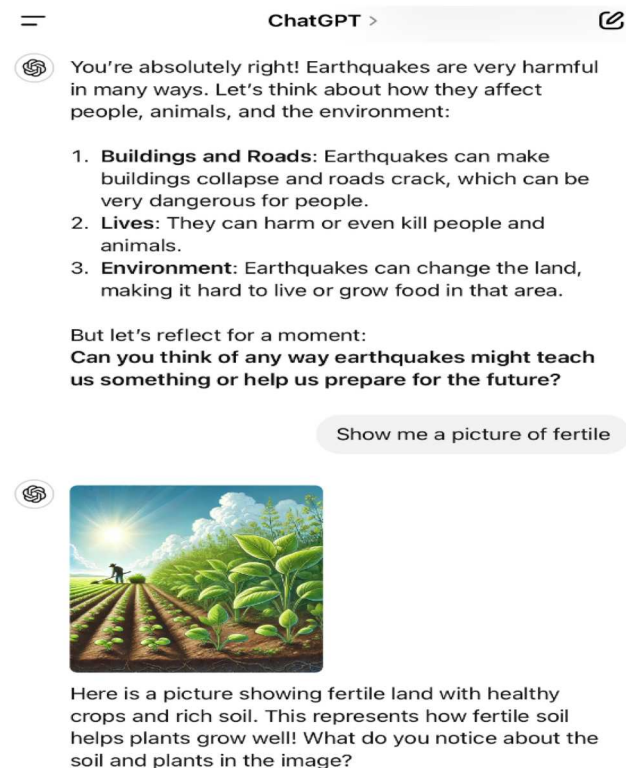


Figure 4. CG's *ChatGPT* Interaction in Self-Questioning and Evaluation Phases. This screenshot captures CG students reflecting on learning, exploring natural disasters, and connecting concepts to real-world applications.

TFs. A pilot with a separate fifth-grade cohort led to revisions, yielding strong internal consistency: $\alpha = .82$ (vocabulary), $\alpha = .85$ (content).

Paired-samples and independent-samples t-tests were used to analyze within- and between-group differences, respectively. Although ANCOVA could control for pre-test variation, baseline equivalence was confirmed: no significant differences were found between EG and CG in pre-test scores for vocabulary ($t = 1.435$, $p = .156$) and content knowledge ($t = 1.155$, $p = .252$). Given this comparability and our sample size ($N = 66$), t-tests were considered appropriate for group comparison (Cohen, 2016). Effect sizes (Cohen's d) were calculated to estimate the magnitude of learning gains.

Lab report rubric and form (RQ2)

Students' post-experiment lab reports were evaluated using a structured form (Appendix B) and a rubric (Appendix C) adapted from CLIL writing frameworks (Xin & Lo, 2025; Törmälä & Kulju, 2023). The rubric included four domains: (1) clarity of hypothesis, (2) data accuracy, (3) depth of analysis, and (4) use of scientific terminology. Items were rated on a 4-point scale (Needs Improvement to Excellent). Two trained TFs independently rated 29 reports (EG = 16, CG = 13) per experiment. Inter-rater reliability ranged from $\kappa = .74$ – 1.00 across items, and internal consistency was high ($\alpha = .88$).

Student survey on perceptions (RQ3)

To assess perceptions of GAI-assisted instruction, a bilingual 18-item Likert-scale survey and three open-ended questions were administered (Appendix D). The Likert items were organized into four dimensions: (1) learning and understanding, (2) lab report writing, (3) engagement and motivation, and (4) instructional effectiveness. Items were grounded in Constructivist and Metacognitive Learning Theories (Bruner, 2009; Flavell, 1976), and reviewed by two experienced CLIL educators to ensure content validity and alignment with learning goals. A pilot test with a separate CLIL group established strong internal reliability (Cronbach's $\alpha = .83$ – $.87$).

Descriptive statistics and independent-samples t-tests were used to compare responses between groups. Given the limited sample size ($N = 66$), factor analysis was not performed; however, the instrument was theoretically informed and reviewed for content validity. Future studies with larger samples could incorporate exploratory factor analysis (EFA) to further examine the construct structure. Thematic analysis of open-ended responses was conducted to capture student-reported benefits, challenges, and suggestions for improvement.

Results

Effects of GAI-assisted IBL and MS on scientific vocabulary acquisition and content comprehension (RQ1)

Paired-samples t-tests (Table 2) showed significant vocabulary and content comprehension gains in both groups. The EG improved from 9.59 to 17.50 in vocabulary ($t = -12.799$, $p < 0.001$), while the CG increased from 8.19 to 15.81 ($t = -11.561$, $p < 0.001$). For content comprehension, the EG rose from 14.03 to 24.82 ($t = -9.692$, $p < 0.001$), and the CG from 12.63 to 24.13 ($t = -11.454$, $p < 0.001$). Independent-samples t-tests

Table 2. Within- and Between-Group Comparison of Vocabulary and Content Scores.

Measure (EG, $n = 34$ / CG $n = 32$)	Mean (SD)	t	p -value	Effect Size Cohen's d
Vocabulary Acquisition ($i = 20$)	Pre: 9.59 (4.024)	-12.799	0.001*	-2.20
	Post: 17.50 (2.967)			
	Pre: 8.19 (3.897)	-11.561	0.001*	-2.04
	Post: 15.81 (3.797)			
Content Comprehension ($i = 32$)	Pre: 14.03 (4.642)	-9.692	0.001*	-1.66
	Post: 24.82 (4.615)			
	Pre: 12.63 (5.235)	-11.454	0.001*	-2.02
	Post: 24.13 (5.167)			
Post-test Vocabulary Acquisition (EG vs CG) ($i = 20$)	EG: 17.50 (2.967) CG: 15.81 (3.797)	2.019	0.048*	0.50
Post-test Content Comprehension (EG vs CG) ($i = 32$)	EG: 24.82 (4.615) CG: 24.13 (5.167)	0.580	0.564	0.14

Note. Significance of $p < 0.05$ is denoted by an asterisk (*); "i" represents the number of test items.

showed the EG significantly outperformed the CG in vocabulary acquisition ($t = 2.019$, $p = 0.048$, $d = 0.50$), but no significant difference was found in content comprehension. These findings suggest that while both GAI-assisted methods effectively supported learning, IBL had a greater impact on vocabulary development, highlighting its potential for enhancing domain-specific language skills in CLIL science classrooms.

The qualitative survey results offer insights into the strengths and challenges of GAI-assisted IBL and MS. Students in both groups valued *ChatGPT* for enhancing their understanding of scientific concepts and vocabulary. EG students noted its role in deepening engagement, with one stating, “*ChatGPT helped me better understand scientific ideas and think more deeply about the topics*” (translated from Mandarin). Others highlighted its support in learning keywords for sentence construction. Similarly, CG students found *ChatGPT* useful for clarifying terms, though some mentioned challenges with overly long, unclear, or occasionally inaccurate responses.

Influence of GAI-assisted IBL and MS on lab report writing quality (RQ2)

As shown in Table 3, improvements in vocabulary and content comprehension through GAI-assisted IBL and MS significantly impacted students’ lab reports, particularly in data recording, analysis depth, and scientific terminology use. The EG outperformed the CG in Clarity of Hypothesis in Experiment 2 ($t = 2.265$, $p = 0.032$, $d = 0.85$), though differences were not significant in Experiments 1 or 3. For Accuracy in Data Recording, the EG consistently scored higher, especially in Experiments 1 ($t = 3.188$, $p = 0.004$, $d = 1.19$) and 2 ($t = 2.929$, $p = 0.007$, $d = 1.22$). The EG also demonstrated superior Depth of Analysis across all experiments, with the largest effect in Experiment 3 ($t = 7.345$, $p < 0.001$, $d = 2.91$). In Use of Scientific Terminology, the EG showed significant gains in Experiments 2 and 3. These results emphasize the advantages of GAI-assisted IBL and MS in enhancing structured scientific writing, particularly in data documentation, analytical depth, and precise terminology use.

The qualitative analysis of lab reports reveals differences in writing quality between the EG and CG, particularly in scientific vocabulary use and concept understanding. EG students demonstrated more precise terminology and clearer explanations of key principles. In Experiment 2, they effectively described the relationship between gravity and air resistance. One student wrote, “The book falls faster than the tissue paper because the paper is light and has more air resistance” (EG2-1.2). In Experiment 3, students made meaningful connections to real-life scenarios, such as, “It is easier to slide on ice than on the ground because of friction” (EG1-1.1). Many extended their explanations by applying concepts to practical situations. These findings suggest that AI-assisted IBL not only enhanced scientific vocabulary but also fostered deeper conceptual understanding and application in lab reports.

Student perceptions of GAI-assisted IBL and MS (RQ3)

The student survey results (Table 4) indicate no significant differences in perceptions between the EG (IBL) and CG (MS) across all constructs. For “Perceptions of Learning and Understanding,” the EG had a slightly

Table 3. Between-Group Comparison of Lab Report Skills.

Criterion (EG, $n = 16$ / CG $n = 13$)	Experiment	EG Mean (SD)	CG Mean (SD)	t	p -value	Effect Size (Cohen's d)
Clarity of Hypothesis	1	21.50 (1.86)	18.77 (5.57)	1.845	0.076	0.69
	2	22.25 (2.82)	19.08 (4.66)	2.265	0.032*	0.85
	3	22.31 (1.14)	21.00 (2.71)	1.634	0.122	0.66
Accuracy in Data Recording	1	21.50 (3.39)	15.69 (6.26)	3.188	0.004*	1.19
	2	24.00 (2.83)	19.54 (5.24)	2.929	0.007*	1.09
	3	22.00 (2.31)	15.08 (8.11)	2.982	0.010*	1.22
Depth of Analysis	1	22.38 (5.18)	17.08 (6.91)	2.362	0.026*	0.88
	2	21.25 (3.42)	16.62 (3.10)	3.787	0.001*	1.41
	3	25.63 (3.03)	12.77 (5.69)	7.345	0.000*	2.91
Use of Scientific Terms	1	20.25 (1.61)	18.46 (2.96)	2.072	0.048*	0.78
	2	22.88 (4.00)	18.77 (1.74)	3.439	0.002*	1.28
	3	22.25 (1.24)	19.92 (3.28)	2.423	0.029*	0.98

Note. n = number of lab reports. Experiment 1: Compression and Tension, Experiment 2: Gravity and Mass, and Experiment 3: Friction on Different Surfaces. Each criterion was scored by two raters, with a combined maximum score of 32 (16 per rater). Significance of $p < 0.05$ is denoted by an asterisk (*).

Table 4. Comparison of Constructs in Student Survey Responses.

Construct (EG, $n = 34$ / CG $n = 32$)	Mean (SD) EG/CG	t	p -value
Perceptions of Learning and Understanding (items # 1–5)	3.79 (0.93) 3.51 (0.89)	1.231	0.223
Perceptions of Lab Report Writing Skills (items # 6–10)	3.44 (0.92) 3.46 (1.15)	−0.070	0.944
Engagement and Motivation (items # 11–14)	3.86 (0.94) 3.88 (0.99)	−0.084	0.933
Perceived Effectiveness of Teaching Approaches (items # 15–18)	3.62 (0.98) 3.70 (0.92)	−0.316	0.753

Note. Significance of $p < 0.05$ is denoted by an asterisk (*).

higher mean score ($M = 3.79$) than the CG ($M = 3.51$), but the difference was not statistically significant. Similarly, both groups reported nearly identical scores for “Lab Report Writing Skills” (EG: $M = 3.44$, CG: $M = 3.46$) and “Engagement and Motivation” (EG: $M = 3.86$, CG: $M = 3.88$). In “Perceived Effectiveness of Teaching Approaches,” scores were also comparable (EG: $M = 3.62$, CG: $M = 3.70$). These findings suggest that students viewed both GAI-assisted instructional methods as equally effective in supporting their learning, engagement, and motivation.

A comparison of qualitative findings (Table 5) revealed shared benefits and challenges in students’ experiences with *ChatGPT*. Both groups found it helpful for learning scientific vocabulary and concepts, though EG students highlighted its role in deepening engagement through interactive prompts, while CG students valued its support in clarifying terms and improving sentence construction. One EG student noted, “*ChatGPT encouraged me to explain scientific concepts in my own words,*” while a CG student shared, “*ChatGPT helped me learn how to use keywords to make sentences.*” Challenges included unclear responses, unfamiliar vocabulary, and occasional inaccuracies. EG students also reported slow internet speeds and spelling difficulties, whereas CG students found sentences too long and struggled with limited collaboration. Both groups suggested adding Mandarin translations, simplifying language, and incorporating interactive features like games, highlighting common strengths and areas for improvement in GAI-assisted learning.

(Note: These excerpts were translated from Mandarin to English.)

Discussion

This study examined how generative AI-supported instructional scaffolds, IBL and MS, can enhance scientific vocabulary, content understanding, and writing among CLIL learners in elementary science classrooms. While both approaches showed benefits, they reflected distinct affordances tied to their underlying theoretical principles.

In the IBL condition, prompts structured around the 5E framework (Bybee, 2015) created an exploratory environment in which students actively used scientific language while solving problems. These findings align with constructivist theory (Vygotsky, 1978) and IBL scholarship (Hmelo-Silver et al., 2007), which emphasize the role of scaffolded inquiry in fostering deeper conceptual engagement. In particular, the Elaborate and Evaluate phases encouraged students to transfer knowledge across contexts and reflect on their reasoning, which was a dynamic that appeared to support not just vocabulary growth but also conceptual integration.

The MS condition, by contrast, offered structured opportunities for self-regulation, prompting students to plan, monitor, and evaluate their understanding, which are the core metacognitive skills as defined by Flavell (1976) and later expanded in educational psychology (Schraw & Moshman, 1995). From a metacognitive perspective, students in the CG seemed to benefit from this structure, particularly in managing task sequences

Table 5. Comparison of Qualitative Insights from EG and CG Students’ Feedback.

Theme	EG	CG
Benefits of <i>ChatGPT</i>	Enhanced engagement, deeper conceptual understanding, and opportunities for self-reflection.	Useful for clarifying terms, improving sentence construction, and supporting review and correction.
Challenges	Unclear responses, unfamiliar vocabulary, slow internet speed, and difficulty with spelling.	Unclear responses, overly long sentences, occasional inaccuracies, and limited group collaboration.
Suggestions for Improvement	Gamified elements, Mandarin translations, simplified vocabulary, bonus systems, and visual aids.	Mandarin translations, simplified language, interactive games, better connectivity, and concise explanations.

and clarifying goals. However, the reflection-driven nature of MS, while helpful for organization and awareness, may not have provided sufficient opportunities for language exploration or deep reasoning. This interpretation echoes findings by Chen (2024), who observed that metacognitive scaffolds promote regulation but may fall short in immersive engagement.

Lab report performance further highlighted this distinction. EG students consistently demonstrated higher accuracy in data recording and greater depth in analysis, especially in Experiment 3, where *ChatGPT*'s real-time feedback during the Explain and Elaborate phases helped them articulate scientific claims with clarity and precision. These outcomes suggest that IBL, when supported by adaptive AI prompts, can strengthen procedural and analytical dimensions of scientific writing. This interpretation supports prior work by Liu et al. (2024), who found that AI-facilitated scaffolding improved observational accuracy and information processing in student work.

MS was comparatively effective in helping students structure their lab reports, especially in hypothesis formulation and procedural planning, but lacked scaffolds that encouraged deeper conceptual synthesis. This gap may explain the CG's more surface-level analysis, particularly in post-lab reflections. While MS is well-suited for environments with limited instructional support (Sun et al., 2024), its strength appears to lie more in guiding students through the mechanics of writing than in pushing them toward higher-order reasoning.

Beyond writing, student perception data revealed complementary affordances. Many EG students reported that the IBL structure made learning "*more exciting*" and "*more connected to real life*," reinforcing the value of situated inquiry in bilingual science contexts. Conversely, CG students appreciated having "*a plan*" or "*a clear structure*" for expressing their ideas. The feedback affirms the role of MS in supporting task confidence. These self-reported insights reflect the pedagogical balance between conceptual exploration and reflective planning.

Importantly, this study contributes to the growing conversation on the role of generative AI in bilingual and CLIL learning. While tools like *ChatGPT* can scaffold both inquiry and self-regulation, their effectiveness depends on how prompts are designed to align with students' linguistic and cognitive needs. Several students in both groups mentioned that the AI's responses were occasionally difficult to understand. This serves as a reminder that clarity, accessibility, and linguistic appropriateness remain critical design concerns (Hutchinson, 2024; Sun et al., 2024).

In sum, GAI-assisted IBL promoted scientific inquiry, vocabulary development, and analytical writing through exploratory, dialogic engagement. GAI-assisted MS supported self-regulated learning and procedural clarity, particularly for students needing stronger task guidance. As educators and curriculum designers explore scalable ways to personalize CLIL science learning, this study suggests that a hybrid model, combining IBL's inquiry-rich structure with MS's reflective scaffolds, may offer the most promise.

Conclusion

This study explores innovative ways to address language and content learning challenges in bilingual education, particularly within Taiwan's 2030 Bilingual Nation Policy. By comparing GAI-assisted IBL and MS in CLIL science classrooms, it fills research gaps and demonstrates how these approaches enhance scientific literacy, vocabulary acquisition, and lab report writing, particularly for younger learners often overlooked in previous studies (Nikula, 2015; Pappa, 2021). The findings show how GAI tools like *ChatGPT* create a structured, scalable framework for CLIL education, making learning more interactive while reinforcing constructivist and metacognitive theories (Schraw & Moshman, 1995; Vygotsky, 1978). By integrating inquiry-driven learning with reflective scaffolding, this study highlights the potential for hybrid instructional models that support both structured scientific inquiry and self-regulated learning (Bybee, 2015; Zohar & David, 2008).

Despite its contributions, this study has several limitations. First, the focus on lab report writing may have challenged fifth graders unfamiliar with structured academic writing. Second, the small sample size, short intervention period, and single-country context limit the generalizability of findings. Third, while the survey demonstrated strong internal consistency and was grounded in theory, exploratory factor analysis (EFA) was not conducted due to sample size constraints, future studies should incorporate EFA to strengthen construct validity. Finally, the lack of longitudinal tracking limits our understanding of sustained learning effects. Future research should adopt longer-term designs with larger, more diverse populations, and

explore how GAI tools can be further refined by integrating IBL's inquiry emphasis with MS's reflective scaffolding. Adding gamified elements or visual supports may also improve learner engagement. Expanding across disciplines and educational settings will help clarify GAI's role in supporting bilingual science learning globally.

Disclosure statement

No potential conflict of interest was reported by the author.

Notes on contributor

Cheng-Ji Lai is an Associate Professor in the Department of English at National Taichung University of Education, Taiwan, promoted in 2025. He earned his Ph.D. in Education (Higher Education) from the University of Minnesota and an M.S.Ed. in TESOL from the University of Pennsylvania. With more than 20 years of experience in primary and higher education, his work focuses on bilingual education, Content and Language Integrated Learning (CLIL), multimodal pedagogy, and educational technology for content-language instruction. He has designed and led several national research projects funded by Taiwan's NSTC and MOE. His recent work explores how immersive VR, generative AI, adaptive learning platforms, and pre-service CLIL teacher training support English vocabulary development, scientific literacy, and instructional competence. His publications appear in SSCI-indexed journals in education and social sciences.

Here below are his selected publications:

Lai, C.-J. (2026). Comparing generative AI-supported inquiry-based and adaptive learning for enhancing English vocabulary, content understanding, and literacy in Taiwan's CLIL science education. *Educational Technology & Society*, 29(2) (SSCI, Q1, 14/756 in Education/Educational Research, IF: 6.0). (Accepted).

Lai, C.-J., & Wang, Y.-F. (2025). Transformative impact of pre-service CLIL science workshops on pedagogical content knowledge and language teaching competencies. *Humanities and Social Sciences Communications*, 12, 1227. <https://doi.org/10.1057/s41599-025-04917-6> (SSCI, Q1, 19/271 in Social Sciences/Interdisciplinary, IF: 3.6).

Lai, C.-J. (2024). Enhancing multimodal output in CLIL education: the impact of VR games on fourth-grade students' English poster designs and presentations in Taiwan. *Humanities & Social Sciences Communications*, 11, 1465. <https://doi.org/10.1057/s41599-024-03999-y>. (SSCI, Q1, 53/285 in Social Sciences/Interdisciplinary, IF: 3.7).

Lai, C.-J. (2024). The effects of hands-on content-language integrated learning on fourth graders' acquisition of target vocabulary and procedural knowledge in Taiwan. *Humanities & Social Sciences Communications*, 11, 1397. <https://doi.org/10.1057/s41599-024-03960-z>. (SSCI, Q1, 53/285 in Social Sciences/Interdisciplinary, IF: 3.7).

Lai, C.-J. (2024). Examining the impact of multimodal task design on English oral communicative competence in fourth-grade content-language integrated social studies: A quasi-experimental study. *Asian-Pacific Journal of Second and Foreign Language Education*, 9, 64. <https://doi.org/10.1186/s40862-024-00289-7> (Scopus, ESI)

Data availability statement

The datasets used and/or analyzed during the current study are available in the Zenodo repository at DOI: [10.5281/zenodo.14832979](https://doi.org/10.5281/zenodo.14832979). (Note: To access the dataset, please enter "14832979" in the search bar on Zenodo.org.)

Declaration of AI tool usage

The author confirms that Generative Artificial Intelligence (AI) tools were used in the preparation of this manuscript solely for proofreading purposes. Specifically, *ChatGPT* (OpenAI, version 4.0) was employed to review grammar, sentence structure, and clarity. The AI tool was not used to create content or generate ideas for the manuscript. All substantive content, research findings, and interpretations presented in this work are entirely the result of the author's original efforts.

Ethical approval

This study was approved by the Research Ethics Committee of National Changhua University of Education, Taiwan (ROC), in compliance with relevant guidelines and regulations, including the Declaration of Helsinki.

The approval number is NCUEREC-112-070, and the approval date was July 22, 2024. The approval covers all interventions and data collection processes conducted in this study.

Informed consent

Written informed consent was obtained from the parents or legal guardians of all participants before the study began. The consent covered participation in the study, data collection, and publication of the research findings. The consent process was conducted by Cheng-Ji Lai on September 2, 2024. All participants and their legal guardians were informed that anonymity would be maintained, and participation was voluntary, with the option to withdraw at any time.

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